UPS SUPPLY CHAIN SOLUTIONS®

Integration 102:
Your Blueprint for Success

Understanding and Preparing to Integrate with UPS
(The advanced guide)

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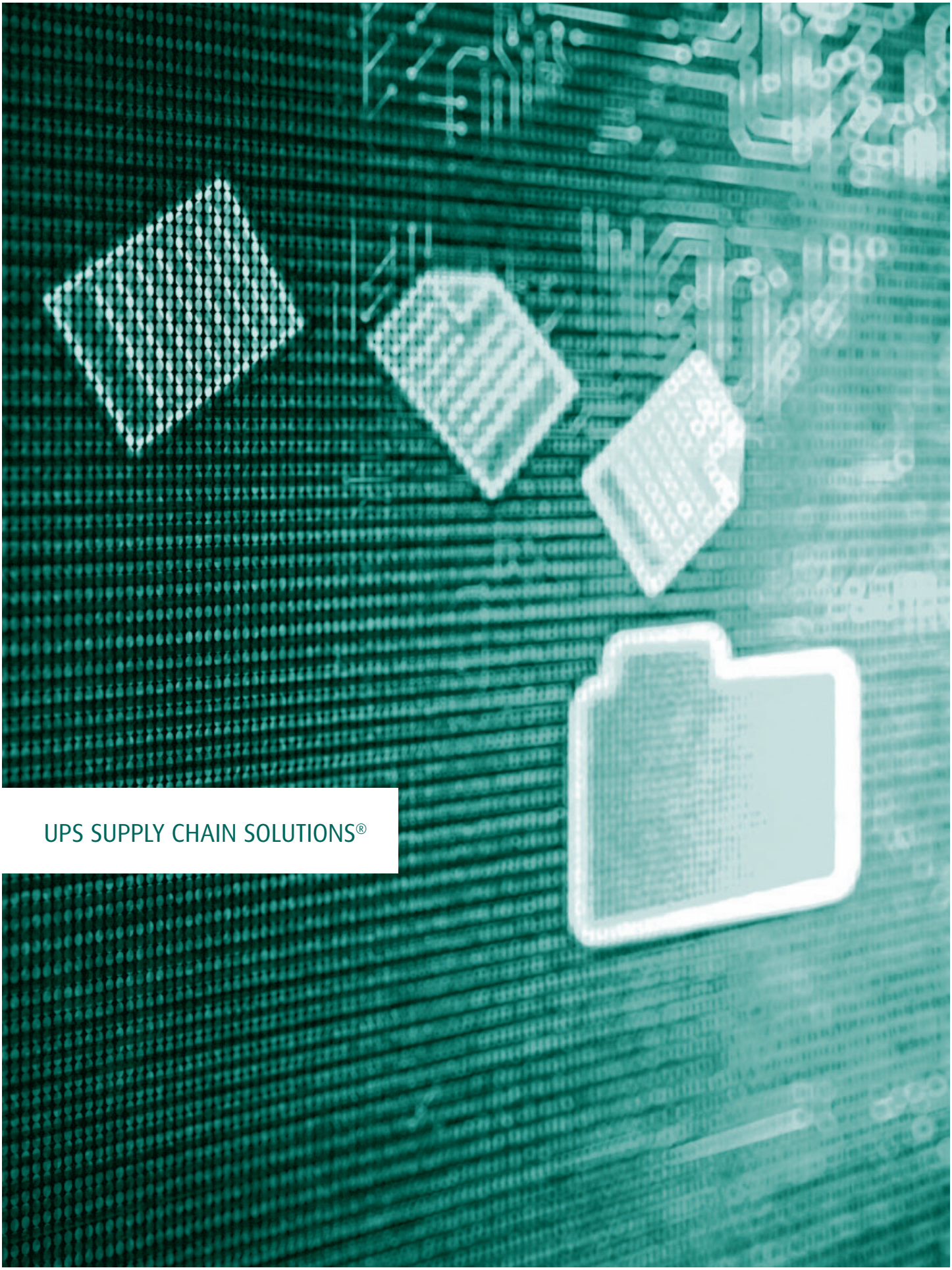

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Table of Contents

Introduction: Welcome to Integration with UPS 2

Chapter 1: Be Prepared (Analysis). 3

Chapter 2: Build a Strong Foundation (Development) 6

Chapter 3: Inspect Your Foundation Closely (Test) 7

Chapter 4: Have a Walk-Through (Go Live). 8

Chapter 5: Facilitate a Smooth Transition (Training and Support) 9

Appendix

- Appendix A: EDI X12 Message Listing 10
- Appendix B: EDIFACT Message Listing 12
- Appendix C: RosettaNet PIP Message Listing 13
- Appendix D: Custom Messages. 14
- Appendix E: Common Integration Terms 15
- Appendix F: Things to Think About 18
- Appendix G: Integration Overview 20

Introduction: Welcome to Integration with UPS

You've made the decision to outsource to UPS. As partners we need to ensure that we can successfully send business information back and forth between our two companies. If you have never traded data with another company before, or have little experience in this area then you might want to refer to our Integration 101 guide for beginners where things are explained in a little more detail. If, however, you are a seasoned veteran when it comes to trading business documents electronically with another company then this guide might be for you. This guide is for reference purposes only and is not intended to be an authoritative work of authorship.

Past experience has shown that project issues and delays often arise during integration testing and go live, usually when both parties are not in sync with the data maps and their contents and/or the field definitions. This guide is intended to help both of us avoid those issues. The guide is written with the intention that whether you work in IT or any other area of the business you will be able to take advantage of the information provided.

The individual chapters are considered the blueprint that will guide us both and ensure we are approaching integration from the same perspective. This will help to ensure we are taking the same approach and communicating in a clear and consistent way. This allows us both to have a positive outcome when we turn on our systems, or flip the switch, so to speak. Like any good blueprint you are going to have to do some preparation and involve people who are considered experts when it comes to your own internal data and systems.

You may already have these answers but just in case you don't, you'll have to ask yourself questions like: "What data do I need to share?" "Who can send it?" "Do I need to receive any data back?" "What system is the data going into?" "Will it work with my system?" "Which operational people need to be involved and do those people know enough about data integration?" "Will they be able to answer all the right 'technical' questions?"

Data integration can be tricky (and technical) but as a seasoned veteran you know that it doesn't have to be. However, if there are terms you want clarified, or you have questions in regard to different formats available, refer to Appendix G, which contains an "integration" overview.

Let's begin.

Chapter 1: Be Prepared (Analysis)

Be Prepared

When dealing with data integration you should create your blueprint. You most likely know what data you want to send and receive. If you don't, then you should make sure you have a solid understanding of the data you want to exchange with UPS. You need to *know your data* or at least have access to someone who does so they can do the proper up front analysis.

Before you start, if you do not have the appropriate level of technical or system knowledge, you may want to make certain that you have someone with such knowledge to be responsible for understanding what fields in each document will need to be extracted or loaded into your systems.

Once you have identified the appropriate individual responsible for the data, you should determine what your blueprint will look like. Below are some basic questions that you or your knowledgeable employee may need to ask:

Data you need to send to UPS

- What data does UPS need in order to manage your business? UPS will tell you what we need from you and which fields are absolutely mandatory and which ones are optional.

Data you need from UPS

- At UPS we have a lot of information that is important to the day-to-day management of your business. Some people are tempted to ask for everything we have rather than just what they need. Past experience shows that when this happens, the company often ends up either not integrating the information into their business application or increasing the development time necessary to build programs, which can impact go live dates. The choice is yours as we can send as much or as little data as you need. Our recommendation would be that in order to determine if it is a piece of data you need, identify what business information the data you are requesting satisfies.

Data characteristics and impacts

- With your data you need to know:
  - **Field lengths for each piece of data.** You should make sure the data will fit into the field it is being sent or loaded into. If you send or receive data that is longer than the field allows, then a portion of the field will be deleted or dropped and the incomplete information will not work for you or UPS.
  - **Product characteristic of each piece of data.** Is the data numeric only, alpha only or a combination of both (alphanumeric)?
  - **Location of the information within your systems.** There may be multiple places where the data will be displayed.
    - Determine what documents the data will print on and where on the document it will appear.

  - Determine if the data shows up on any reports. If it does, then identify which reports and where. Also identify if it is the same field name on the documents and reports as it is within the application. There are times where a field name has a secondary reference that is used on reports so understanding the location and field name used on the document or report may be beneficial. A suggestion may be to make a list of all fields, where they show up within the various business applications within your system, what documents they appear on and what reports they need to be shown on. Your list should include application, document and report names, as well as program names for each. The list can then be provided to the teams doing the development and testing as a quick reference.
- **New data fields required.** If you are requesting a piece of data that you've never received into your system before, you should understand what it is and what its purpose is. For example, you should identify:
  - Where and what business applications it appears in
  - What documents it needs to be printed on
  - What reports it needs to show up on

  You may want to make a list for the developers and testers, and if possible, include screen, document and report samples. This may assist the developer to better understand what you want and how you want it to appear.
- **Field functions.** It is important that both parties understand what a field is called so we can compare apples to apples and not apples to bananas. You might call a piece of data within your application a "sales invoice" but at UPS, it might be called a "sales order number."
- **Field dependencies.** Identify if any other fields are dependent on the information contained within the field being reviewed. It is important to know this information because if the data you are populating is incorrect and another field is dependent on it (like a customer shipment tracking program) then it could cause a problem for you. If you aren't sure, ask the question. If the answer is yes, ensure you cover off a testing scenario during the build phase. We'll talk more about testing scenarios in a little bit.

By doing the analysis and being prepared you may be able to avoid any unnecessary additional expenses that happen when rework is required due to misunderstanding. When people don't clearly communicate and aren't on the same page with each other, the time needed to implement the solution lengthens and the project cost increases. In addition you avoid unnecessary frustration for all parties involved.

Once the previously addressed issues have been resolved, then you are ready for the next step: the data mapping session. This is when both of us get together to agree on what data is going to be sent back and forth, what the data will be used for, where in the file each piece of data will reside and whether or not it is required (mandatory) or informational (optional).

Data Mapping

Data mapping can make or break a project. It is a key part of the initial analysis. This is really the detailed version of the blueprint. It includes the layout and design for each file being sent and/or received. It is where we will identify and agree to the data being sent back and forth between our two companies and:

- Identify what the name of the data is,
- Identify what the field length and characteristics of the data are,
- Identify where each piece of data will go within the system it is being sent to, and
- Identify what the data will be used for.

Both of us should invest the time to get together and agree on the data maps, either in a face-to-face meeting or via conference calls. During these meetings or conference calls, we should each strive to:

- **Invest enough time to go through the mapping discussion.** Our people will need to go through the documents being sent or received, field by field, to ensure the information is available and can be sent. Depending on how many documents you want to send back and forth, the length of time will vary.
- **Have the right people at the meeting.** You may need a technical expert who understands your data and business systems in order to provide the correct information. Otherwise you may end up with an "apples to bananas" scenario and misinformation being provided.
- **Understand the timing of when files are sent/received.** Try to understand when each document type will be sent and if there are important cut-off deadlines. If all orders to be shipped that day need to be received by UPS by a specific time, then your system will need to send it prior to that cut-off time if you want your orders to be sent on time.
- **Agree on the naming conventions.** Strive to ensure that if you refer to a field by more than one name, you know what those alternate names are and can relate that information to UPS.

We encourage you to make sure you have taken the time to do the necessary planning so that you can hand over your blueprints to a developer and trust them to map the programs for loading and sending data in accordance with the agreed-upon specifications.

Also, you should invest the time to prepare and engage the right people to participate in the mapping discussions in order to better ensure the correct information is sent back and forth between yourself and UPS.

Chapter 2: Build a Strong Foundation (Development)

Once you have done the analysis and completed your data maps (blueprint) your development work can begin. We encourage you to make sure your developers build based on the data map and not outside of it. The following are a few things to consider for the development phase:

- **Allocate enough time and resources for development work.** This is where you build the programs that will load data into or extract data out of your system.
  - If you have new information being received from UPS, provide the information (the list you created during analysis) to your developer. This is the document that identifies where the information needs to show up within the various business applications in your system, what documents it needs to show up on and where it should appear on the reports.
- **Ensure resources are knowledgeable about your system.** If your employees did not participate during the data mapping (analysis) you may want to make sure those employees have access to the people who were involved during the data mapping discussions in case they have questions or need clarification.
  - If you don't have people with the correct expertise to do this, let us know as we may be able to either provide resources or identify a third-party company to help you out.
- **Perform unit testing.** After you build each program to extract data or input data, review the changes and test them. Make sure that all information is going into the correct fields as expected.
- **Build or provide test scripts.** Work with UPS to identify and document the testing requirements that will be used. This helps prepare both of us for user acceptance testing.
  - Identify how many documents will be run through testing.
  - Identify the desired outcome for each test.
    - If you don't have experience building test case scenarios, ask us. We've done many of these and may be able to suggest what standard testing scenarios to use according to document type. It may be more convenient for you to review and modify a standard testing scenario rather than creating one from scratch. Either way, you should make sure you get what you need to test on both sides.

Chapter 3: Inspect Your Foundation Closely (Test)

We need to test what we build to ensure it will work as expected when we go live. Unlike Chapter 2 where you test each piece as you build it (unit test), now both of us need to test everything together. This is the place where the connectivity testing (system integration or SIT) and data testing (user acceptance testing or UAT) occur.

Here are our recommendations to help you prepare for SIT and UAT testing:

- Identify and agree to the testing requirements in advance (during your build phase).
- Plan to do the connectivity testing (SIT) as a separate step so we can ensure that our two systems can talk to each other as expected. Do not blend SIT with UAT. Think of it as turning on a light to make sure there is power being received into the switch.
- Ensure all of your information and business scenarios are accounted for and checked off during UAT.
- During UAT, limit the number of scenarios to two or three.
- Ensure there is someone accountable for reviewing and signing off on each document type you send and/or receive. This is like a building inspector in our analogy.
  - Note: If you assign someone who is not familiar with the documents or your various in-house screens that users see, they most likely can do the testing but it will take them longer to verify the information.
- Track and record issues. Expect to find a few errors and try not to panic when they occur. The whole point of testing is to flush out any problems before we go into a production environment.

Chapter 4: Have a Walk-Through (Go Live)

It is important that before we "go live" and move 100 percent into production mode, we send a few transactions through to ensure things are functioning as expected.

Here are a few things to review before fully sending and receiving all business transactions:

- **Align dates with UPS.** Make sure we are both ready to start on the same date and that neither of us had any problems with the code promotion from a test environment to a production environment.
- **Run a "golden" transaction through the system for each transaction type.** Make sure you can send and receive data and that the data goes to the correct spot in the production environment for each transaction.
- **Monitor the flow of activity.** Continue to monitor the flow of activity over the next several days in order to ensure that everything is sending and receiving as expected.

By controlling the flow of data at "go live" you are able to catch and address any errors that may have been missed during testing. It is much easier to back out one transaction and correct an issue than it is to back out a hundred or a thousand transactions. Making sure everything is working as expected should provide everyone with peace of mind and allows for data to flow back and forth freely and correctly.

Chapter 5: Facilitate a Smooth Transition (Training and Support)

It is important to ensure that the right people have been trained, know what to expect and are aware that you are moving from a manual to an electronic format as their processes may slightly change. Below are some suggestions to keep in mind.

- **Train people in advance of the "go live."** This allows everyone to properly be prepared and makes "go live" run more smoothly.
- **Ensure that the right support processes are in place.** Once you "go live" there may be small issues that arise. Your people will have to be able to know where to go to get help and have people trained who can support them.
- **Train the support staff.** Make sure your support staff are fully trained in order to understand the types of questions they are being asked. They should not only understand what the problem is but should know how to get it resolved for your operational team.
- **Get sign-off on training.** Obtaining sign-off from the users who do the testing may help to confirm that the users are satisfied with the results and ready to proceed.

Stay engaged until all open issues are resolved. Ensure all issues, even if they are only training issues, are resolved and that everyone is satisfied things are working properly.

Appendix A: EDI X12 Message Listing

The following is a list of most commonly used EDI messages. Note that UPS SCS has the capability to handle additional messages if required:

101	Name & Address Lists
110	Air Freight Details & Invoice
180	Return Merchandise Authorization and Notification
204	Motor Carrier Load Tender
210	Motor Carrier Freight Details & Invoice
211	Motor Carrier Bill of Lading
214	Transportation Carrier Shipment Status Message
240	Motor Carrier Package Status
304	Shipping Instructions
310	Ocean Freight Receipt & Invoice
315	Ocean Status Details
317	Delivery/Pickup Order
754	Routing Instructions
810	Invoice/Commercial Invoice
812	Credit/Debit Adjustment
820	Payment Order/Remittance Advice
823	Lockbox
824	Application Advice
830	Planning Schedule and Release Capability
832	Price/Sales Catalog
838	Trading Partner Profile
844	Debit/Credit Adjustment
845	Price Authorization Acknowledgement/Status
846	Inventory Inquiry/Advice
849	Response to Product Transfer Account Adjustment
850	Purchase Order
852	Product Activity Data
855	Purchase Order Acknowledgement
856	Ship Notice Manifest/ASN
860	Purchase Order Change Request — Buyer Initiated
861	Receiving Advice/Acceptance Certificate
867	Product Transfer and Resale Report

Appendix A: EDI X12 Message Listing (cont.)

875	Grocery Products Purchase Order
880	Grocery Products Invoice
888	Item Maintenance
940	Warehouse Shipping Order
941	Inventory Status Report
943	Warehouse Stock Transfer/Shipment Advice
944	Warehouse Stock Transfer/Receipt Advice
945	Warehouse Shipping Advice
947	Warehouse Inventory Adjustment Advice
990	Response to a Load Tender
997	Functional Acknowledgement

Appendix B: EDIFACT Message Listing

The following is a list of most commonly used EDIFACT messages. Note that UPS SCS has the capability to handle additional messages if required:

CUSDEC	Customs Declaration
CUSRES	Customs Response
IFCSUM	Forwarding & Consolidation Summary
IFTMIN	Instruction Message
IFTSTA	International Multimodal Status Report
INVOIC	Invoice

Appendix C: RosettaNet PIP Message Listing

The following is a list of most commonly used RosettaNet PIP messages. Note that UPS SCS has the capability to handle additional messages if required:

2A1	Item Master/Distribution New Product Information
3B2	ASN/Notify of Advance Shipment
3B12	Order Request/Request Shipping Order
3B12 ACK	Order Request Acknowledgement
3B13	Notify of Shipping Order Confirmation
3B18	Notify of Shipment Documentation
3B3	Shipment/Status/Distribute Shipment Status/Proof of Delivery
3C3	Notify of Invoice
4B2	Receipt Confirm/Notify of Shipment Receipt
4C1	Inventory Snapshot/Inventory Adjustment /Distribute Inventory Report

Appendix D: Custom Messages

While it is advised to stay away from customization, UPS SCS does have the ability to handle the following 9Z series custom messages:

9Z1	Inventory Snapshot
9Z2	Shipment Order
9Z3	Order Confirm/Accept/Cancel
9Z4	Ship Confirm
9Z5	ASN
9Z6	Receipt Advice/PO Receipt Advice

Appendix E: Common Integration Terms

Acronym	Description
3PL	3rd Party Logistics
A2A	Application to Application
A2C	Application to Customer
ACK	Acknowledgement
AIF	Application Interface File
ANSI	American National Standards Institute
AS2	Transfer Protocol used to send/receive EDI files
ASCII	American Standard Code for Information Interchange
ASN	Advanced Shipment Notice
B2B	Business to Business
B2C	Business to Consumer
CGI	Common Gateway Interface
CSI	Common System Interface
CTCP	Client-to-Client Protocol
DBMS	Database Management System
DFD	Data Flow Diagram
DHTML	Dynamic HTML
DIMS	Dimensions
DISA	Data Interchange Standards Association
DMA	Data Mapping Analysis
DNS	Domain Name Service
DTD	Document Type Definition
EDI	Electronic Data Interchange
EDIA	Electronic Data Interchange Association
EDIFACT	Electronic Data Interchange for Administration, Commerce and Trade
EFT	Electronic Funds Transfer
EMS	EDI Management System
EOF	End of File
EOM	End of Message
ETS	Electronic Trading System
FA	Functional Acknowledgement
FAQ	Frequently Asked Question
FEDI	Financial Electronic Data Interchange
FTP	File Transfer Protocol
GIC	Global Integration Center
GIGO	Garbage In, Garbage Out

Appendix E: Common Integration Terms (cont.)

Acronym	Description
GPS	Global Positioning System
GRE	Global Rules Engine
GUI	Graphical User Interface
HTML	Hyper Text Markup Language
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
I/O	Input/Output
IE	Internet Explorer
IP	Internet Protocol/Intellectual Property/Inner Pack
IPX	Internet Packet Exchange
ISA	Industry Standard Architecture
ISDN	Integrated Service Digital Network
ISO	International Standards Organization
ISP	Internet Service Provider
Kbps	Kilobits per second
LAN	Local Area Network
MAPI	Messaging Application Programming Interface
MB	Megabyte
MBps	Megabytes per second
Mbps	Megabits per second
NACK	Negative ACKnowledgement
NAT	Network Address Translation
NDIS	Network Driver Interface Specification
NFS	Network File System
NIC	Network Interface Controller
NTP	Network Time Protocol
ODBC	Object Data Base Control/Open Data Base Connectivity
ODETTE	Organization for Data Exchange by Tele Transmission in Europe
PAEB	Pan American EDIFACT Board
PAN	Personal Area Network
PAP	Password Authentication Protocol
PC	Personal Computer
PD	Public Domain
PDA	Personal Digital Assistant
PDF	Portable Document Format
PDML	Product Data Markup Language
PING	Packet INternet Groper

Appendix E: Common Integration Terms (cont.)

Acronym	Description
PIP	Partner Interface Protocol
PPP	Point-to-Point Protocol
PPTP	Point-to-Point Tunneling Protocol
RAM	Random Access Memory
RAS	Remote Access Service
RF	Radio Frequency
RFID	Radio Frequency Identification
RNIF	RosettaNet Implementation Framework
SFTP	Secure FTP
SKU	Stock Keeping Unit
SMTP	Simple Mail Transport Protocol
SNA	Systems Network Architecture
SNA/DS	System Network Architectural Distribution Services
SNMP	Simple Network Management Protocol
TCP	Transmission Control Protocol/Trade Chain Partner (Cdn CSA)
TCPIP	Transmission Control Protocol — Internet Protocol
TP	Trading Partner
UAT	User Acceptance Testing
UDF	User Defined Fields
UN/EDIFACT	United Nations rules for Electronic Data Interchange for Administration, Commerce and Transport
UNECE	United Nations Economic Commission for Europe
UNTDID	United Nations Trade Data Interchange Directory
UOM	Unit of Measure
UPC	Universal Product Code/Units Per Carton
VAN	Value Added Network
VPN	Virtual Private Network
WAN	Wide Area Network
WAP	Wireless Access Point/Wireless Application Protocol
WLAN	Wireless Local Area Network
WPA	Wi-Fi Protected Access
WPAN	Wireless Personal Area Network
WWAN	Wireless Wide Area Network
WWW	World Wide Web
XML	Extensible Markup Language
XMTP	XML Mail Transport Protocol
XSD	Extensible Schema Definition

Appendix F: Things to Think About

There are a few things that you should consider when building any plans for integration. Below is the list of the most common questions:

"Why do I need to know when my system blackout periods are? I don't even know if I have any."

- Blackout periods are those times when your IT department closes the system to any program changes (code freeze) in order to maintain system stability. When programs are changed, there is always a risk, no matter how minimal, that the change may affect another program adversely. Blackout periods are usually identified in advance by the IT department and normally occur around high peak period events throughout the year. It is important to know when your own system will be locked out for code changes, and when UPS may have blackout periods. The last thing you want is to find out right before you are ready to load the changes and "go live" that your system has a code freeze and you have to wait two or three weeks until it is over to continue.

"I've heard that sometimes people 'go live' and have all kinds of problems. Will that happen to us?"

- Usually the projects that have problems are the ones where corners were cut and one of the parties did not properly prepare for the "go live." No matter how much testing you do in a test environment it does not always guarantee that your production environment is 100 percent the same and everything will go through without a hitch. While we should both strive to move into production without any systemic problems it is important to understand the risk. The recommendation is that before opening the floodgates into your system, you should send one or two files through as a "golden transaction"' and make sure it goes all the way through the system without any issues. If there is an issue, you can quickly back out the code and fix the problem without having to delete a large number of files. Once you've successfully sent a few files all the way through, you can then let the files flow.

"Why do I/my people have to go through such a detailed mapping session?"

- It is important to ensure that both companies are defining fields the same way (oranges to oranges) and that field sizes are the same length. Many times companies want to skim over this piece of the process and those are the companies who have problems with data not showing up correctly in their system.

"What is the big deal with having special characters in our fields?"

- Special characters are the characters that generally appear at the top of a keyboard above the numbers (*, &, %, etc). Many companies use some of the characters as part of their company name, address and sometimes product names. Depending on the character used, it may or may not be an issue. With integration, all pieces of information (data) must be separated so that the program knows when one field starts and when it stops. Some special characters are the triggers for programs to read the data and understand the purpose of each piece of data, based on where it is located in the file, in order to place it in the correct spot within the system application that it is being loaded into. The most common separator used is the asterisk (*) but sometimes other characters can be used. If you use an asterisk in your name (i.e., ABC * Company) then the system will put ABC in the first field and then Company in the other field. We recommend that you ask your UPS project manager what special characters you need to avoid so you don't run into these types of issues.

"I want to duplicate everything I do today — can I?"

- You can always duplicate what you do; however, one advantage of integration is that you can automate your system functions. You have the ability to reduce labor by eliminating unnecessary data entry and can provide access to information quicker. Look for opportunities to streamline your processes in order to maximize efficiency when automating processes.

"If my data isn't 100 percent correct can you fix it?"

- If you will always be missing information or know that a certain piece of information always needs to be corrected, UPS may be able to put a process in place to accommodate you. The challenge is that this becomes a manual process. When you automate you want to try and eliminate the manual touch points as much as possible because people may make mistakes.

"What happens if I use FTP? Will you guarantee my data security somehow?"

- No. Most trading partners will ask you to use secure FTP; however, they will normally accept FTP as long as you do not hold them accountable for in-transit data. Many companies will ask you to sign an agreement that releases them from the security of data while it is in transit. This is often referred to as a "hold harmless" agreement. UPS will require a hold harmless agreement if you do decide to use non-secured FTP.

Appendix G: Integration Overview

The following overview is meant for those with little knowledge on the subject. If you are comfortable in an area you can skip over it or skim through to the areas where you need some clarification. The core of this guide is in the chapters. The overview is to help you, or your employees, better understand integration as a whole.

Data Integration

When two companies work together there usually is a need for each of them to share common business information (aka data) back and forth. This can be done through old fashioned methods like faxing, mailing or couriering documents, but a faster way to share business information is to send the data back and forth between the two companies electronically. The party that data is sent to receives it electronically and then loads, or "integrates," the data they receive into their internal system(s). Those are the essentials of UPS data integration: the ability of our two companies to share and utilize business information back and forth.

Trading Partners

We are now trading partners since you will be trading data with UPS electronically. The information you trade back and forth depends on your business and the types of documents that you need to exchange. Some companies will hire an independent third party to manage their electronic data needs. That company would also be considered a trading partner.

Types of Documents Exchanged Electronically

There are many types of documents that can be sent back and forth between our two companies. Below is a listing of the most common. The functions for some vary depending on the services that you've chosen to use with UPS:

- Purchase Order (PO)
- Invoice Sales Order (SO)
- Advanced Shipping Notice (ASN)
- Receipt
- Funds Transfer
- Export Notice
- Import Notice

Once you have decided what types of documents you want to exchange with UPS you must then determine the specific type of formats and method of file transfers that you will use.

Likewise with data integration, both parties must agree to share data using the same format in order for the exchange of data to work properly and to reach the proper destination. There are several different formats that may be used, such as Electronic Data Interchange (EDI), Extensible Markup Language (XML) and Flat File.

Formats for Sending Information

EDI

EDI is structured and very organized, which allows both companies to follow a set format. EDI is the most commonly used format because it is so closely structured. It is a direct method for exchanging documents from one business to another (B2B) or from one application to another application (A2A).

XML

XML is a more flexible way for two companies to create common information formats and share both the format and the data over the Internet or intranet. With XML the structure will remain the same but you have some freedom to work and agree with your trading partner (UPS) on how the data will be set up.

It is important, however, that even though XML messages are easier to read they are typically larger files than EDI messages. This is something that needs to be considered if you are planning on sending high volumes of data back and forth between yourself and UPS. It is also important to note that within XML there are two formats — these are:

- **DTD**, or **Document Type Definition**, is a set of rules (base blueprint) for the fields being used in XML. Technically, it is the definition of the elements, attributes, entities and notations used within XML. When the rules are followed, the DTD is valid, when the rules are not followed, the DTD is invalid.

  The biggest challenge with DTDs is that you need to use a unique structure and most people have to take the time to learn that unique structure and for that reason many companies tend to lean more towards using XSD.

- **XSD,** or **Extensible Schema Documentation**, defines how the data will be stored within that structure. Like DTDs, it is also the base blueprint from which to work, but in addition to the definition of the elements, attributes, entities and notations, it also allows you to tightly define and validate the content of an element or attribute against a data type. This helps to ensure that you get what you need where you need it. This is what makes XSD a more attractive option to some. With XSD you may be better able to define what information is required, where it will be and what it will look like. XSD is UPS's preference over DTD because there is more flexibility, it is easier to understand, data can be validated *and* controls are still maintained.

Flat File

Flat files are data files that do not adhere to any structure. In most cases additional information is required in order to interpret the files such as file format but usually enough information can be provided to automate the process and eliminate a good portion of the manual labor required. The most common flat file structure used is Microsoft Excel. This is because the data can be listed in rows where each row represents a record. Each record has information organized in a specified format or sequence. Some trading partners tend to shy away from using a flat file because quite often there is additional work required; however, that does not mean that you cannot use flat files. Note that if you are missing a lot of data, it may put additional labor requirements on UPS to complete the missing information manually before being able to load it into the system. Yet, if you send files that contain complete data, it can still be an ideal solution when you don't have other EDI options or resources available to you.

Regardless of which format you choose, both of us must agree to use the same format *and* the files must be sent through a common electronic link in order for the exchange of data to be successful. This ensures that we are both going to the same place to send and receive data.

Below are the most commonly used file formats:

Format	Definition	Notes	Messages
EDI ANSI-X12 (also called X12)	U.S. standards under the control of the American National Standards Institute (ANSI).	■ Developed separately from Europe and other foreign countries. ■ Most common version is 4010. ■ Both companies must work from same version.	Refer to Appendix A for a list of common messages.
EDIFACT	The EDIFACT standard (EDI for Administration Commerce and Transport) is used in Europe and other countries and is the only EDI standard that is truly accepted worldwide.	■ Provides standard formats for business documents. ■ Incorporates features that meet international requirements.	Refer to Appendix B for a list of common messages.
RosettaNet	Uses specific processes or PIPs (partner interface processes) to identify common business definitions for all documents (messages) exchanged.	■ Allows trading partners of all sizes to connect electronically to process transactions and move information within their extended supply chains.	Refer to Appendix C for a list of common messages.

UPS is very flexible and can trade data using most any format you may require. The important thing to remember is that both of us must start from the same format, using the same version when applicable to ensure we build a solid blueprint together. This is the basis for the data mapping session that we will have. The data mapping by far is the most important piece of integration because it is where the details of the data that is to be sent and/or received are worked out between our two companies. If we are not using the same formats and have not identified the data details being sent back and forth we both will fail. However, when the data details are reviewed up front and confirmed to be accurate, we will have a better chance to succeed when we are ready to go live.

Translation and Data Mapping

"Translation" is the process of taking documents or files from your computer, your internal application, or provided to you from UPS, and manipulating them into a format that is acceptable to the application or system that it is going to be loaded into. This is where data mapping comes in. The developers must create a layout (blueprint) of the data that is being taken from various places within your system application and put it in the correct spot within the document format you agreed to use with UPS (e.g., the Ansi X12 document format). Not only will you need to map data coming out of your system, you will also need to map and translate incoming documents UPS sends you so that they will be able to be recognized by, and loaded into, your system.

The data mapping session is extremely important as you need to ensure that the right information not only goes to the right fields, but that both you and UPS have exactly the same definition for each field. If you map the information to the wrong spot in your system, your employees and your customers may receive or view bad information. If you map information (data) into the right spot, but it is the wrong information, your employees and your customers could receive or view bad information.

Once you've completed your mapping session and know what is going where and to whom, we need to agree on the method for sending the files back and forth between our two companies. This is how the data is going to get to UPS from your internal systems or to your internal systems from UPS. This is known as a "transfer protocol." The most common question asked in regard to this is "What transfer protocol will you be using?"

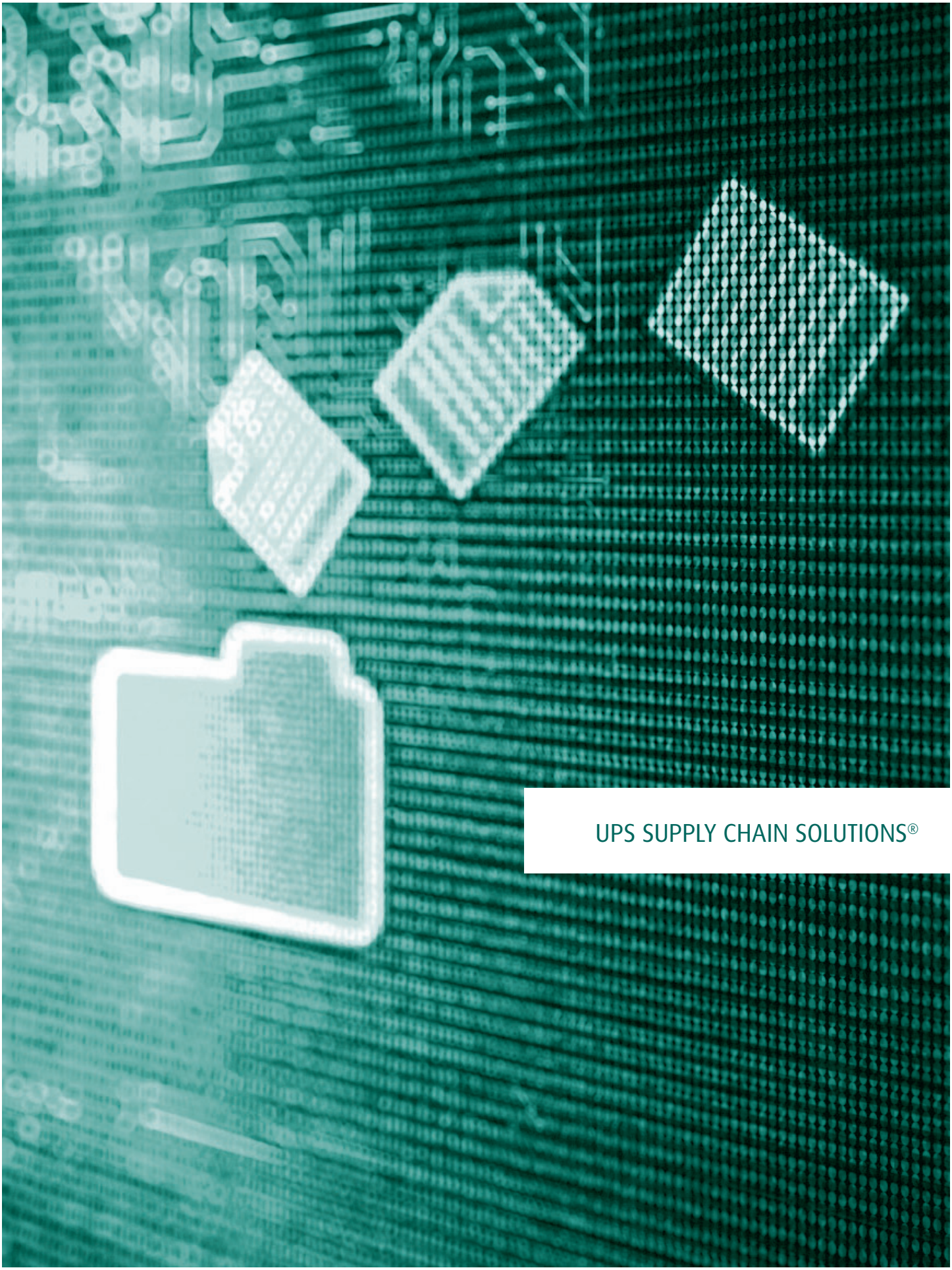
Transfer Protocols

There are different methods ("protocols") for sending and receiving ("transferring") files. This is referred to as "transfer protocols."

The Internet is the most common way that data is sent back and forth between a company and UPS because it is something most people have familiarity with and usually have access to. It is also one of the most economical means of sending data. It is important, however, to note that with the Internet, your data can become public unless it is secured, so choosing the right way to transfer your data may be really important to you. When balancing cost and security, you may want to find both a cost-effective and secure method of moving your data.

The following is a list of the more common file transfer protocols being used today:

Protocol	Definition	Description	Notes
AS2	Applicable Standard 2	Ensures that the proper level of security is maintained when transmitting your information to UPS. Note that AS2 messages are always sent using the HTTP or HTTPs protocol (see below).	■ Originally designed for EDI but can also be used by XML (flexible). ■ Files can be sent right away. ■ Files can be batched (but don't have to be). ■ Messages can be digitally signed (but don't have to be). ■ You can use secure certificates. ■ Similar to proof of delivery, you can confirm the delivery of the message.
HTTP	Hypertext transfer protocol	A different standard of web servers and systems to communicate and transfer files back and forth. Most companies prefer the secure version of HTTP.	■ You initiate the request by establishing a session to a particular port on a server (the server must be "listening" and send back the "ok" before the data is sent). ■ Several contacts back and forth will not harm your system. However, if you open up a port to someone and give access to options that allow them to delete data, place data on your server, etc., you could be exposing yourself to risk if someone other than UPS is able to find and connect through the port.
HTTPs	Hypertext transfer protocol secured	The same as HTTP only it is a secure version that uses encryption and a secure identification network to ensure the safety and security of your data.	■ Data is encrypted to better secure the safety of your information. ■ You need to have the key to decrypt the data. Without the key, you cannot access or understand the data.
RNIF	RosettaNet Implementation Framework	Specifically used to transfer RosettaNet messages between two companies.	■ Cannot be used for any other message type, such as X12 or EDIFACT. ■ Based on HTTP, MIME (multipurpose internal mail extensions) and XML standards and follows the same rules as they do, but is specific to the RosettaNet messages.
FTP	File Transfer Protocol	Transfer protocol used to send data directly between two companies.	■ Data is unsecure and safety cannot be guaranteed. ■ UPS will require you to sign a document confirming that you acknowledge the risk you are taking if you choose this particular protocol method.
sFTP	Secure File Transfer Protocol	Same as FTP only this is the secured version.	■ Data is secured through authentication.



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